Performance Analysis Cheat Sheet

1. Performance Tuning: iterative → profile → optimize → measure → repeat.
   Bottlenecks: memory, compute, workload balance, communication.
2. FLOPS (Floating Point Ops/sec):
   - Theoretical: FLOPS = Sockets × Cores × Clock × FLOPs/cycle
   - Measured: FLOPS = (Ops × FLOPs/iteration) / Time
   Example (CPU i7-970): 166 GFLOPs (SP), 83 GFLOPs (DP).
   Example (GPU RTX 3080): ~29.77 TFLOPs (SP).
3. Bandwidth (GB/s):
   - Formula: Bandwidth = Data Size / Time
   - Host→Device (H2D) vs Device→Host (D2H): usually H2D > D2H.
   - Theoretical < Actual due to overhead, protocol, inefficiencies.
4. Speedup:
   - Ratio: Speedup = T_old / T_new
   - % Improvement = (T_old - T_new) / T_old × 100
5. Amdahl’s Law:
   Formula: S = 1 / [(1 - p) + (p / s)]
   - p = fraction parallelized, s = speedup factor
   - Max speedup = 1 / (1 - p) (even with infinite processors).
   Example: 80% ×2 → 1.67× better than 20% ×8 → 1.21×.
6. Key Takeaways:
   - FLOPS & Bandwidth both matter (compute + memory).
   - Focus on largest runtime portions.
   - Optimize both serial and parallel code.